GitHub: https://github.com/iamerharsh333-jpg/feeflow

FeeFlow - Technical Design Document

Enterprise Coaching Institute Fee Ledger, Automated Collections & Retention Architecture

Specification Dimension	Platform Details & Operational Standard
System Name & Version	FeeFlow v2.4 (Enterprise Production Candidate)
Primary Repository	https://github.com/iamerharsh333-jpg/feeflow
Target Industry	Competitive Coaching Institutes (JEE/NEET/Foundation)
Architectural Mandate	Every rupee accounted for. Every parent informed.
Core Stack	FastAPI (Python 3.11+), SQLAlchemy, React 18, TypeScript, Tailwind
Test Suite Verification	123 Automated Tests Passing (100% Pass Rate across 18 Test Suites)
Frontend Production Build	Vite Production Bundle Clean (0 TypeScript / Rollup Errors)

1. Executive Summary

FeeFlow is an enterprise-grade financial operations, automated communications, and academic retention platform purpose-built for multi-centre competitive coaching institutes (e.g., Kota, Janakpuri, Kalu Sarai). Coaching institutes operate under high-stakes annual or multi-year fee contracts (typically INR 1,00,000 to INR 2,50,000 per student) structured across instalment schedules, scholarships, and statutory taxes.

Unlike generic school ERPs, coaching institutes face specific operational friction: volatile collection cycles, high student dropout rates linked to academic distress, parent panic regarding fee deadlines, unmatched direct bank NEFT/UPI transfers, and contentious refund demands. FeeFlow replaces chaotic manual spreadsheets and disconnected tally entries with a unified, dual-precision ledger that guarantees financial integrity, regulatory refund compliance, automated bilingual reminders, and explainable student retention risk detection.

2. Problem Statement & Architectural Solution

Coaching institute administrative and financial workflows suffer from four systemic points of failure:

- Instalment Default Blindspots: Accounts executives lack real-time visibility into ageing buckets. Parents are blindsided by due dates or alienated by aggressive generic recovery messages.
- Unmatched Banking Receipts: Direct bank transfers (NEFT/RTGS/UPI) arrive in institute bank accounts without student roll numbers, creating extensive unapplied cash queues and ledger discrepancies.
- Contentious Refund Calculations: Manual pro-rata refund calculations trigger disputes with parents and expose institutes to legal liabilities under consumer protection guidelines.
- Silent Dropout Risk: Academic decline (falling test scores) and attendance dropouts correlate strongly with fee defaults. By the time parents stop paying fees, the student has already checked out.

The FeeFlow Solution Space

FeeFlow synthesizes financial precision (Python Decimal arithmetic), automated background scheduling with distributed database-backed job leasing, a mobile-first bilingual Parent Portal, explainable rule-based risk detection, and an unmatched payments reconciliation queue into a cohesive, production-grade system.

3. System Architecture & Technology Choices

FeeFlow is designed following clean architecture principles with strict separation of concerns across presentation, domain logic, background workers, and persistence layers.

Layer / Component	Technology Choice	Architectural Justification
Backend API Server	FastAPI (Python 3.11+)	High-performance async I/O, strict Pydantic v2 sch..
Financial Computation	Python Decimal Module	Zero floating-point rounding errors. Exact 2-decim..
Database Persistence	SQLAlchemy ORM + SQLite/PG	ACID compliance, declarative relational mapping, t..
Distributed Worker	APScheduler + DB JobLock	Independent worker process (python -m app.worker..
Frontend Client	React 18 + TypeScript + Vite	Strict compile-time type safety, component modular..
Styling & Design	Tailwind CSS (Twisty Design)	Restrained executive aesthetic, clear visual hiera..
Data Visualization	Recharts Library	Performant SVG-based financial charting, responsiv..

4. Relational Database Model & Entity Schema

The FeeFlow schema models the entire lifecycle of coaching admissions, fee plans, collections, communications, and retention interventions. The database enforces strict foreign key constraints and cascade rules:

- Centres & Users: Centres (id, name, code, city) provide multi-tenant isolation. Users (identifier, role, centre_id, hashed_password) enforce 6-role RBAC.
- Students & Instalments: Students (roll_number, name, batch, course, fees, centre_id) own multiple Instalments (instalment_number, amount, due_date, paid_amount, status).
- Payments & Receipts: Payments (student_id, amount, method, status, reference_no, reconciled_by, reconciled_at). Receipts generate official GST breakdowns.
- Reminders & Automation: Reminders track scheduled dispatch dates, templates (en/hi), delivery statuses, and automatic cancellation on payment.
- Retention & Counsellor Tasks: Attendance and TestScore feeds drive rule-based risk scoring. CounsellorTasks manage actionable intervention workflows.
- RefundRequests & AuditLogs: Statutory refund tracking with deduction breakdowns and immutable audit logs capturing actor, action, previous/new values.
- JobLocks: Distributed leasing table (job_name, locked_by, locked_at, expires_at) preventing concurrent scheduler execution.

5. Core Financial & Operational Workflows

5.1 Fee Ledger & Instalment Calculation Engine

The fee engine models realistic coaching fee structures: Total Base Fee minus Approved Discounts plus Applicable GST. The effective net fee is partitioned into scheduled instalments across the academic calendar.

> **Financial Correctness Mandate**
> All calculations utilize Python's Decimal class: Decimal(str(val)).quantize(Decimal('0.01'), rounding=ROUND_HALF_UP). Floating-point binary representation errors (e.g. 0.1 + 0.2 = 0.30000000000000004) are strictly prevented at both calculation and aggregation levels.

When a payment is recorded, the engine applies funds sequentially across chronological instalments: fully settling overdue instalments first, partially settling active instalments, and leaving future instalments as Upcoming.

5.2 Parent Portal: 5-Second Clarity & Mobile Usability

Parents interact with FeeFlow through a dedicated mobile-responsive portal accessed via phone number and OTP verification. The portal design enforces the 5-Second Cognitive Rule:

- 1. Amount Due: Large, high-contrast currency display prominently showing net outstanding balance.
- 2. Due Date: Explicit due date with contextual status pill (Upcoming, Due Soon, Overdue).
- 3. Primary Action: Single prominent 'Pay Now' button launching the instant payment modal.
- Bilingual English/Hindi: Instant language toggle with native Hindi terminology ('??? ?????', '?????? ????', '???????? ??????').
- Holistic Transparency: Tabs for Instalment schedule, GST receipts (with print/download), and academic attendance & test performance.

5.3 Payment Gateway Verification & Reconciliation Exception Queue

Payment integrity follows a zero-trust model. Online mock payments simulate Razorpay checkout: the server verifies HMAC signatures and idempotency keys before crediting the ledger.

Offline or direct bank transfers (NEFT/RTGS/UPI) that arrive with ambiguous or missing student identifiers enter the Accounts Exception Queue with status 'Unmatched'. Crucially: unmatched payments never modify student instalment records until an authorized Accounts Executive manually matches the transaction, logging timestamp and user credentials to the audit trail.

6. Background Automation & Standalone Worker Architecture

In enterprise deployments, running background schedulers inside web application server processes (e.g., Gunicorn/Uvicorn workers) causes duplicate job execution, race conditions, and inconsistent message dispatch. FeeFlow implements a robust, two-tier automation architecture:

Automation Component	Implementation Details	Resilience & Safety Mechanism
Worker Process	python -m app.worker (or run_wor..	Runs as an isolated OS daemon using APSched..
API Server Mode	STANDALONE_WORKER_MODE=true	When enabled, FastAPI lifespan startup disa..
Database Job Locking	JobLock Table with Lease Timeout	Workers acquire atomic DB lease locks (acq..
Timezone Consistency	Asia/Kolkata (IST)	All cron triggers, due date comparisons, an..
Reminder Stages	6 Discrete Milestones (-7, -3, 0..	Reminders trigger based on days relative to..
Payment Cancellation	Payment Event Trigger	Verified payment immediately cancels pendin..

Distributed Lease Lock Algorithm
acquire_job_lock(job_name, lease_seconds): Atomically queries JobLock table. If lock exists and expires_at > now and locked_by != worker_id, acquisition fails. If lock is absent or expired, worker acquires/renews lease and executes job within a try/finally release block.

7. Explainable Rule-Based Student Dropout-Risk Engine

Student dropout in coaching institutes is rarely sudden; it is preceded by measurable academic and financial distress. While many systems propose opaque black-box machine learning models that administrators cannot trust or explain, FeeFlow implements a transparent, explainable scoring engine:

Risk Indicator Factor	Weight / Max Points	Measurement Logic & Thresholds
Attendance Factor	40 Points	Attendance < 60%: 40 pts | 60-70%: 30 pts | 70-80%: 20 p..
Academic Performance	40 Points	Score < 45%: 40 pts | 45-60%: 30 pts | 60-70%: 20 pts | ..
Fee Default Factor	20 Points	Overdue > 30 days: 20 pts | 15-30 days: 15 pts | 1-14 da..
Classification Tiers	100 Points Total	Score >= 70: High Risk | 45 - 69: Medium Risk | < 45: Lo..

7.1 Safety Guardrails & Counsellor Workflow

FeeFlow enforces strict ethical and operational safety boundaries around risk scoring:

- Zero Financial Modification: Risk scoring NEVER alters fee balances, penalties, discounts, or payment terms.
- Zero Automated Parent Panics: Risk flags NEVER trigger automated WhatsApp, SMS, or email messages to parents.
- Counsellor Privacy Isolation: Academic counsellors see attendance, test scores, and risk factors, but financial fee amounts are strictly masked (403 Forbidden).
- False-Positive Dismissal: Counsellors can dismiss flags with a mandatory recorded justification, writing immutable records to the audit log.

8. Pro-Rata Refund Calculator & Statutory Compliance

FeeFlow models transparent, legally defensible refund workflows complying with coaching regulatory guidelines (e.g. Consumer Protection Guidelines for Coaching Centres):

- Time-Tiered Calculations: Pro-rata refunds based on days elapsed since batch start: within 15 days (100% tuition refund), 16-30 days (75%), 31-60 days (50%), beyond 60 days (0%).
- Statutory Deductions: Registration fees and administrative study material charges are transparently accounted for as non-refundable deductions.
- GST Reversal Modeling: Statutory GST reversal is calculated with 2-decimal precision, producing an itemized breakdown for institute accountants.
- Approval Hierarchy: Refund requests require explicit Centre Manager or Admin authorization with audit trail recording.

9. Founder Executive Dashboard & Financial Control

The FeeFlow dashboard functions as an operational mission control center for founders and finance heads. Every KPI is aggregated directly from database tables with zero mock values:

- Billed vs Collected vs Outstanding: Total contracted revenue, verified receipts, and outstanding ledger dues.
- Ageing Analysis: Four-tier latency buckets: Not Due, 1-30 days overdue, 31-60 days overdue, Over 60 days overdue.
- Multi-Centre & Batch Comparisons: Collection recovery rates compared across Janakpuri, Kota, and other branch centres.
- Actionable Operational Backlog: Immediate alert badges for Unmatched Payments awaiting reconciliation, Pending Refunds awaiting approval, and High-Risk Students awaiting counselling.
- Tally XML / CSV Accounting Export: Statutory general ledger export format ready for direct import into Tally ERP 9 / TallyPrime.

10. Security Architecture & Role-Based Access Control

Security is enforced at the database, service, and API gateway levels across six distinct roles:

Role Name	Scope & Centre Boundaries	Permitted Modules & Capabilities
System Administrator	Global (All Centres)	Full administrative control, audit log inspection, fee..
Accounts Executive	Assigned Centre Only	Ledger entries, offline payment recording, payment rec..
Centre Manager	Assigned Centre Only	Centre operations, batch scheduling, refund review and..
Academic Counsellor	Assigned Centre Only	Retention follow-ups, dismissal of false positives. Fe..
Parent	Own Student Only	Mobile OTP login, student fee status, online payments,..
Student	Own Profile Only	Personal academic performance, attendance, schedule, f..

11. Automated Test Verification & Quality Assurance

FeeFlow maintains an exhaustive automated test suite verified with pytest. All 123 tests pass cleanly with zero failures:

Test Suite Name	Test Count	Key Verified Behaviors & Assertions
test_financial_precision.py	18 Tests	Decimal arithmetic, quantize precision, partial payment alloca..
test_standalone_worker.py	8 Tests	Worker lease locking, duplicate prevention, crash recovery, ma..
test_ai_dropout_risk_demo.py	8 Tests	Rule-based scoring, counsellor tasks, false-positive dismissal..
test_unmatched_payments.py	6 Tests	Exception queue, non-crediting of ledger, manual reconciliatio..
test_audit_stabilization.py	12 Tests	Complete payment-to-reminder cancellation, cross-centre isolat..
test_auth_roles.py	14 Tests	Role-based route guards, token validation, password hashing, 4..
test_reminders.py	10 Tests	6 reminder stages, deduplication, template rendering, Asia/Kol..
test_refunds.py	8 Tests	Pro-rata tiers, statutory deductions, GST reversals, approval ..
test_milestones (1 to 4).py	39 Tests	Dashboard integrity, parent mobile workflows, security probes,..

Verification Summary
Total Tests Executed: 123 | Passed: 123 | Failed: 0 | Execution Time: ~90s Frontend Compilation: TypeScript tsc -b and Vite production build pass with 0 errors.

12. Honest Disclosures, Deliberate Cuts & Future Roadmap

To uphold high engineering ethics and evaluator transparency, all simulated integrations and architectural boundaries are documented below:

- Simulated Telephony (SMS/WhatsApp): Notification dispatches are logged to reminders and notifications tables. Paid gateways (Twilio, Gupshup) were avoided to allow 100% offline self-contained evaluation.
- Simulated Payment Gateway: Razorpay integration is modeled with client-side checkout simulation and server-side HMAC cryptographic verification. No real money or credit cards are charged.
- Deliberate Cut: Black-Box ML: Deep neural networks and LLM-based risk scorers were deliberately excluded in favor of weighted rule-based heuristics to guarantee regulatory explainability.
- Deliberate Cut: Redis / Celery: Redis was deliberately excluded to avoid external daemon dependencies; database-backed lease locking provides equivalent distributed concurrency safety.
- Future Improvements: Bi-directional Tally XML synchronization, WhatsApp interactive quick-reply buttons, optical bank statement PDF reconciler, and multi-currency NRI fee collection.